Confounders and controls

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Name the intended change. Cross the hidden alternative. Record the controls.

Identify confounding and propose matched/crossed comparisons without claiming a toy sensitivity check proves complete causal reasoning.

Keep one habit: Control contract → method/corpus/query ledger → native four-cell leaf design → grouping/order probes → toy-rule audit → independent timing confound.

MAIN LESSON · SECTION 1

Locate the changes that accompany the intended method difference

> A comparison can have alternative explanations even when its score improves.

Learning goals

I can explain the **mechanism** and terminology of **confounder**s and **control**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve fair baselines, grouping and study conditions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Experimental units and independence. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **confounder**s and **control**s using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **confounder**s and **control**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Define the intended factor and the conditions that need control

> A whole-system comparison is valid to describe, but cannot silently isolate one changed component.

Begin with a method and question-set change

Imagine changing both the retrieval method and the questions used to judge it. If the score changes, which change explains it? A control plan holds relevant conditions fixed or varies them systematically so the intended comparison is interpretable. Document the corpus, cases, cutoff, scoring, environment and order.

Matching a few recorded fields helps, but it cannot prove that every alternative explanation was removed.

Define confounding and control

A **confounder** changes alongside the intended factor and can offer another explanation. **Control**s aim to isolate the proposed **mechanism**.

Complete main and picture design contract

Control-plan item	Complete fictional classroom contract
Intended main factor	Retrieval method A versus B
Main unwanted changes	A uses corpus c1 and Q1–Q3; B uses c2 and Q4–Q6
Fixed declared fields	Cutoff 3, rubric r1 and toy-local environment
Proposed matched comparison	B uses c1 and Q1–Q3 as A does, with method B retained
Evidence gap	No retrieval outcomes or real rerun are supplied; matching metadata is a plan/check, not a performance result
Separate picture factor	Leaf condition healthy/spotted; nuisance background white/blue
Original coupled cells	Healthy on white; spotted on blue only
Crossed design	Healthy-white, healthy-blue, spotted-white, spotted-blue
Scope	Fictional design metadata and an explicit toy paper-only rule; no actual images, collection, classifier training or causal experiment

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace method, corpus and cases as separate changes

> A claim about the method needs a design that does not change its evidence at the same time.

Original matched-input and order-control procedure

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Complete intended/unintended-change ledger

Added main metadata field	A original condition	B original condition	Comparison interpretation
Method	A	B	Intended difference
Corpus	c1	c2	Unwanted difference for isolated method question
Queries	Q1, Q2, Q3	Q4, Q5, Q6	Unwanted case difference
Cutoff/rubric/environment	3 / r1 / toy-local	3 / r1 / toy-local	Declared matches, not independent execution verification
Proposed repair	Keep A as stated	Method B with c1/Q1–Q3/3/r1/toy-local	Matched-plan proposal; no new score supplied
Remaining evidence	Actual outcomes, order and further conditions	Still need to be collected/recorded	Do not claim all confounding solved

State which conditions and outcomes remain missing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Confounders and controls — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
New method uses better corpus and different queries while old method uses the original set.	For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.	Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict more data with the same coupled condition pairs

> More rows cannot fill a condition cell that was never observed.

Current task and response

- New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict healthy-white and spotted-blue only again

- Change: New cases still use only healthy-white and spotted-blue.

▼ Check after predicting

Prediction: The collection does not resolve the background confounding.

Reason: The same two variables remain coupled instead of being varied separately.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read the four cells and the possible paper-colour shortcut

> Changing the background diagnostically is different from proving a full causal mechanism.

Main corpus/query confound and checked repair

New method uses better corpus and different queries while old method uses the original set.

Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate leaf condition and controlled-view scope

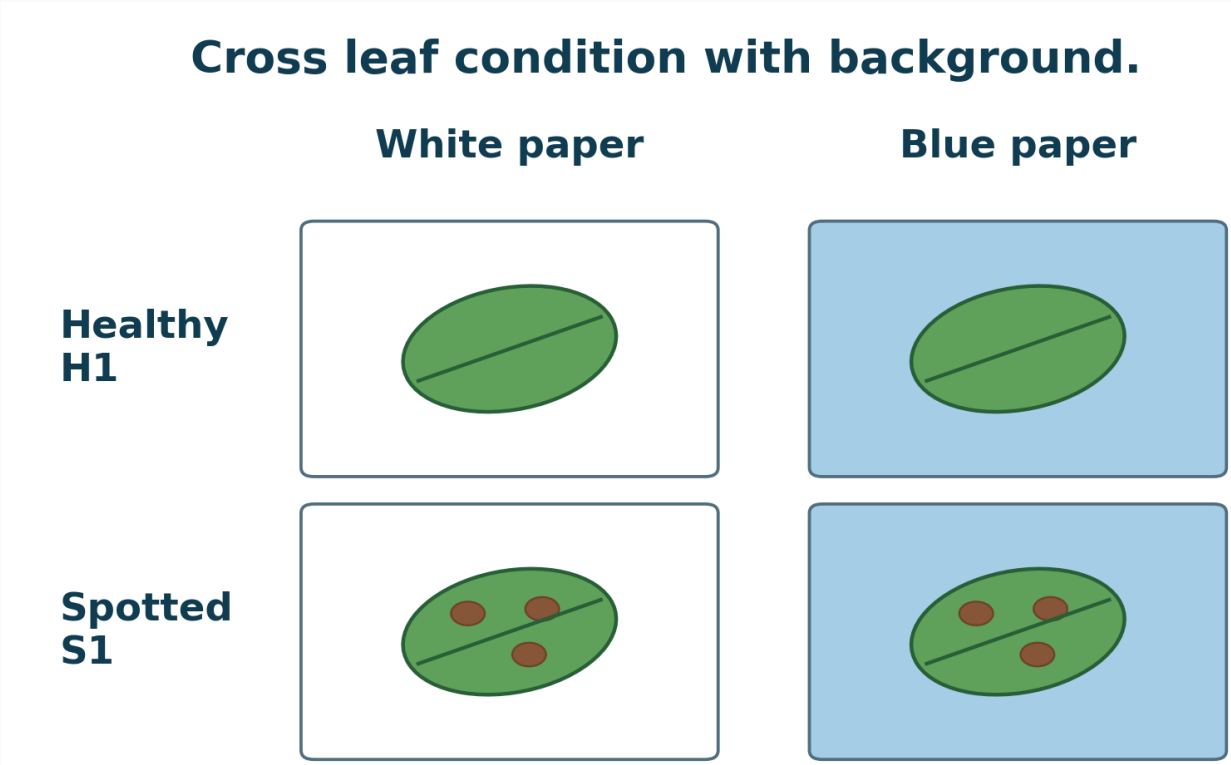
Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.

Goal is leaf condition, not paper colour. A proposed check uses both paper colours for each condition.

The picture is a separate leaf-background problem. If every healthy leaf is on white paper and every spotted leaf on blue, a rule using paper colour alone can fit those observed pairs. That possibility does not establish what a real model actually uses. Cross the two conditions and backgrounds, and inspect a paired test where the same held-out leaf changes background while its condition stays fixed.

The native illustration shows healthy H1 and spotted S1 in both colours; these are repeated views of two objects. For a new-plant evaluation, keep all views of those objects out of training. A changed-background response can show sensitivity, not the model's complete internal reasoning or universal behavior.

See all four combinations in the native illustration



AI ACADEMY / AI-7/10 · 02 · TRACE THE CASE

Follow the deciding steps



01 Identify coupled variables

Condition and background change together in the original collection.



02 Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.



03 Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify coupled variables	Condition and background change together in the original collection.
2. Cross the conditions	Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.
3. Inspect the shortcut	Compare predictions when the same leaf's background changes under a controlled test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Complete crossed-cell and toy-rule ledger

Separate condition/background cell	Original collection	Crossed check	Explicit paper-only prediction
Healthy / white	Present	Included	Healthy: correct
Healthy / blue	Missing	Added	Spotted: incorrect
Spotted / white	Missing	Added	Healthy: incorrect
Spotted / blue	Present	Included	Spotted: correct

Separate condition/background cell	Original collection	Crossed check	Explicit paper-only prediction
Toy-rule total	2/2 correct on coupled records	2/4 correct on crossed records	Constructed alternative explanation, not actual model behavior
Object count	H1 and S1	Same H1/S1 in both backgrounds	Four records from two leaf objects

Predict collecting only the original coupled cells

- Change one thing: New cases still use only healthy-white and spotted-blue. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-7/10 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Compare predictions when the same leaf's background changes under a controlled test.

Changed condition

New cases still use only healthy-white and spotted-blue.

Worked changed-case result

The collection does not resolve the background confounding.
Why: The same two variables remain coupled instead of being varied separately.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain sensitivity and unchanged confounding

- A changed-background test can diagnose sensitivity; it does not prove the model’s complete internal reasoning.

- Read the labelled picture: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

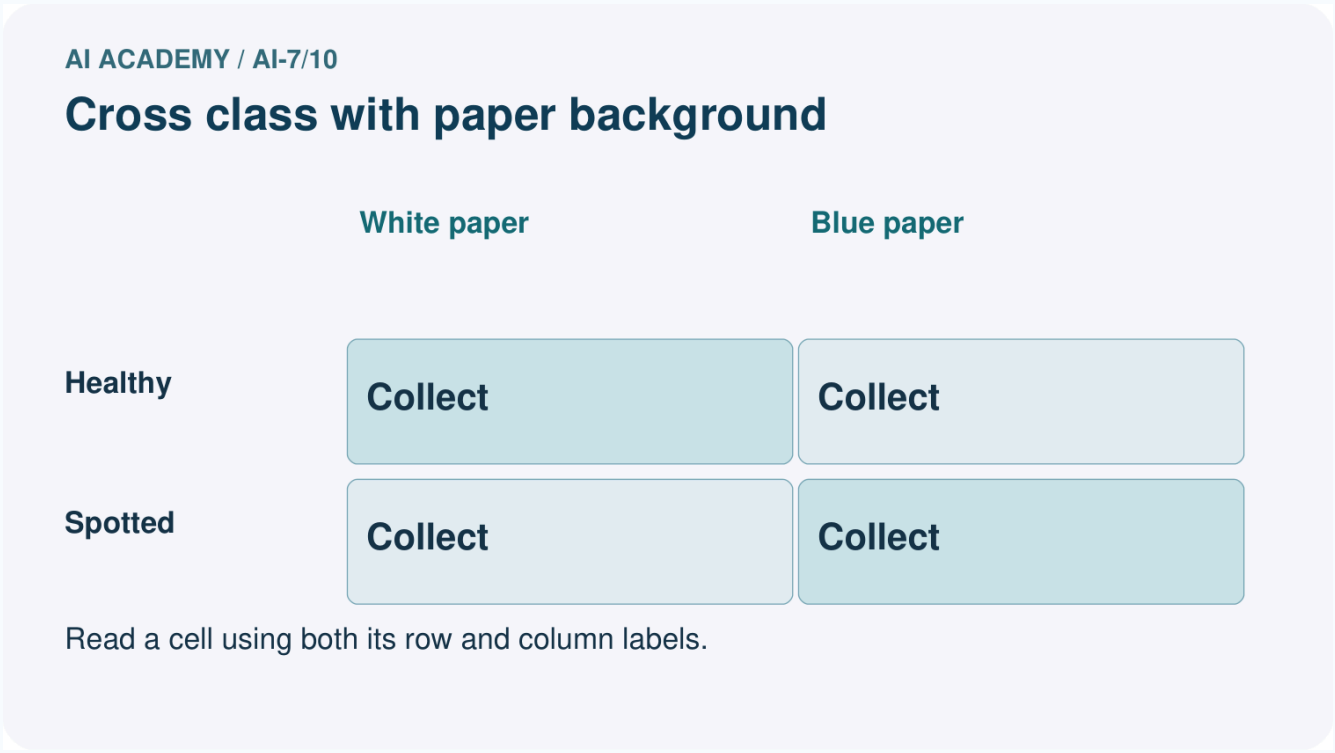
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.
</details>

A pictorial worked case: Confounders and controls

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/10

Confounders and controls

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify coupled variables

Condition and background change together in the original collection.

2. Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.

3. Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

CHANGE ONE THING

New cases still use only healthy-white and spotted-blue.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit what a control removes and what it leaves untested

> A better design narrows alternative explanations without guaranteeing every causal claim.

Core confounding qualification and failure analysis

Controls address specified alternatives, not every possible unknown **confounder**.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete design-cell, grouping and order probes

Confounding/control probe	Expected interpretation
Method, corpus and query set all change	Cannot isolate the method effect from this comparison
Only listed metadata match	Useful check; actual execution and unrecorded conditions may still differ
Healthy-white and spotted-blue only	Condition and background remain coupled
More of the same two cells	Missing cross-combinations remain missing
All four cells included	Improves this design; still inspect object grouping, sampling and other differences
Paper-only rule fits original records	Alternative explanation is possible; not proof a real model uses it
Same held-out leaf changes prediction with background	Shows sensitivity under that test, not complete reasoning
Repeated views counted as independent plants	Wrong unit count; retain object identity

Confounding/control probe	Expected interpretation
Order/time changes with method	Randomize or counterbalance relevant order/time conditions and record them

Guided repair and explanation

- A report omits this qualification: **Control**s address specified alternatives, not every possible unknown **confounder**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise naming the intended factor and a plausible alternative

> Propose a specific matched or crossed design, not a vague request for more data.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example "**Confounder**s and **control**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Confounder**s and **control**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

My explanation using a pictured step and the given facts

Reserve the selected method/time-of-day confound

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run a complete control-record and crossed-condition investigation

> Inspect missing cells and the known toy rule rather than imagining a live model result.

Current multi-change comparison investigation

New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Runnable declared-condition and toy-rule audit

```
"""Declared comparison controls and a deliberately paper-only toy rule."""
from collections import Counter
from copy import deepcopy
A={'method':'A','corpus':'c1','queries':
['Q1','Q2','Q3'],'cutoff':3,'rubric':'r1','environment':'toy-local'}
B={'method':'B','corpus':'c2','queries':
['Q4','Q5','Q6'],'cutoff':3,'rubric':'r1','environment':'toy-local'}
def unintended_changes(a,b):
 fields={'method','corpus','queries','cutoff','rubric','environment'}
 if type(a) is not dict or type(b) is not dict or set(a)!=fields or set(b)!=fields:
  raise ValueError('complete declared control records required')
 return [field for field in ['corpus','queries','cutoff','rubric','environment'] if
a[field]!=b[field]]
main_changes=unintended_changes(A,B)
matched_B={**deepcopy(A),'method':'B'}
assert main_changes==['corpus','queries'] and unintended_changes(A,matched_B)==[]
# Separate picture metadata: H1 and S1 recur; these four rows are not four leaf objects.
original=[{'leaf_id':'H1','condition':'healthy','paper':'white'},
          {'leaf_id':'S1','condition':'spotted','paper':'blue'}]
crossed=[{'leaf_id':leaf,'condition':condition,'paper':paper}
         for leaf,condition in [('H1','healthy'),('S1','spotted')] for paper in
['white','blue']]
COMBINATIONS={(condition,paper) for condition in ['healthy','spotted'] for paper in
['white','blue']}
def design_cells(rows):
 if type(rows) is not list or not rows:raise ValueError('nonempty supplied metadata
required')
 cells=Counter()
 for row in rows:
  if type(row) is not dict or set(row)!={'leaf_id','condition','paper'}:
   raise ValueError('exact leaf/condition/paper metadata required')
  if type(row['leaf_id']) is not str or not row['leaf_id'] or
```

```
(row['condition'],row['paper']) not in COMBINATIONS:
   raise ValueError('declared leaf and cell values required')
  cells[(row['condition'],row['paper'])]+=1
 return {'cells':dict(cells),'missing_cells':sorted(COMBINATIONS-set(cells)),
         'records':len(rows),'leaf_ids':len({r['leaf_id'] for r in rows})}
def paper_only_rule(paper):
 if paper not in ('white','blue'):raise ValueError('known paper fixture required')
 return 'healthy' if paper=='white' else 'spotted'
def evaluate_rule(rows):
 design_cells(rows)
 return [{'leaf_id':r['leaf_id'],'condition':r['condition'],'paper':r['paper'],

'prediction':paper_only_rule(r['paper']),'correct':paper_only_rule(r['paper'])==r['condition']}
for r in rows]
before=deepcopy(crossed)
original_cells=design_cells(original);crossed_cells=design_cells(crossed)
original_results=evaluate_rule(original);crossed_results=evaluate_rule(crossed)
assert len(original_cells['missing_cells'])==2 and crossed_cells['missing_cells']==[]
assert sum(r['correct'] for r in original_results)==2 and sum(r['correct'] for r in
crossed_results)==2
assert len(crossed_results)==4 and crossed_cells['leaf_ids']==2 and crossed==before
print('Main unintended declared changes:',main_changes,'matched-design
fields:',unintended_changes(A,matched_B))
print('Original versus crossed cells:',original_cells,crossed_cells)
print('Deliberately paper-only rule on coupled records:',original_results)
print('Same toy rule on crossed records:',crossed_results)
print('The rule fits original2/2 yet only2/4 crossed, illustrating an alternative
explanation. It does not reveal what an actual trained model uses internally.')
print('All conditions/labels are supplied fictional metadata. No photos, collection,
classifier training, randomization, network timings or real causal experiment is performed.
Matching listed fields does not prove all confounders controlled.')
```

Expected mismatches, cells and evidence limits

1. Predict the main corpus/query mismatches before running or tracing the metadata audit. The proposed matched B removes those listed differences, not every possible confounder, and no performance result is computed for it. Separate leaf fixtures show two original cells and all four crossed cells.

   The intentionally paper-only rule scores two of two original records but two of four crossed records, while only two leaf IDs exist. This illustrates an alternative explanation without training or diagnosing any real classifier. No images, randomization, new collection or network timing run occurs. Keep actual local output separate from a paper trace and from the independent time-of-day design task.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: method is coupled with time of day

> Repair the comparison and record the timing conditions.

Complete selected independent task

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save intended factors, controls and unexplained differences

> Keep diagnostic sensitivity and broad causal explanations separate.

Project task and artifact

Add a labelled **confounder**s and **control**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **confounder**s and **control**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local metadata run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the alternative explanation and a design that could challenge it

> A control plan must say what changes and what stays comparable.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **confounder**s and **control**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Confounder	A differing condition offering another explanation
Control	A design constraint preserving comparison conditions
Counterbalancing	Varying order to reduce systematic order differences

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: **Control**s address specified alternatives, not every possible unknown **confounder**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Confounders and controls — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
New method uses better corpus and different queries while old method uses the original set.	For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.	Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.

By the end, I can	How I will show it
Explain the mechanism and terminology of confounders and controls	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.
- Which labelled step matches this detail? Compare **prediction**s when the same leaf's background changes under a **control**led test.
- 1. Identify coupled variables
- 2. Inspect the shortcut
- 3. Cross the conditions
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect the shortcut Compare **prediction**s when the same leaf's background changes under a **control**led test.

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Changed-case question and optional hint

- Changed-case practice: New cases still use only healthy-white and spotted-blue. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The collection does not resolve the background confounding.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The collection does not resolve the background confounding. The same two variables remain coupled instead of being varied separately.

Repair and retain the record

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

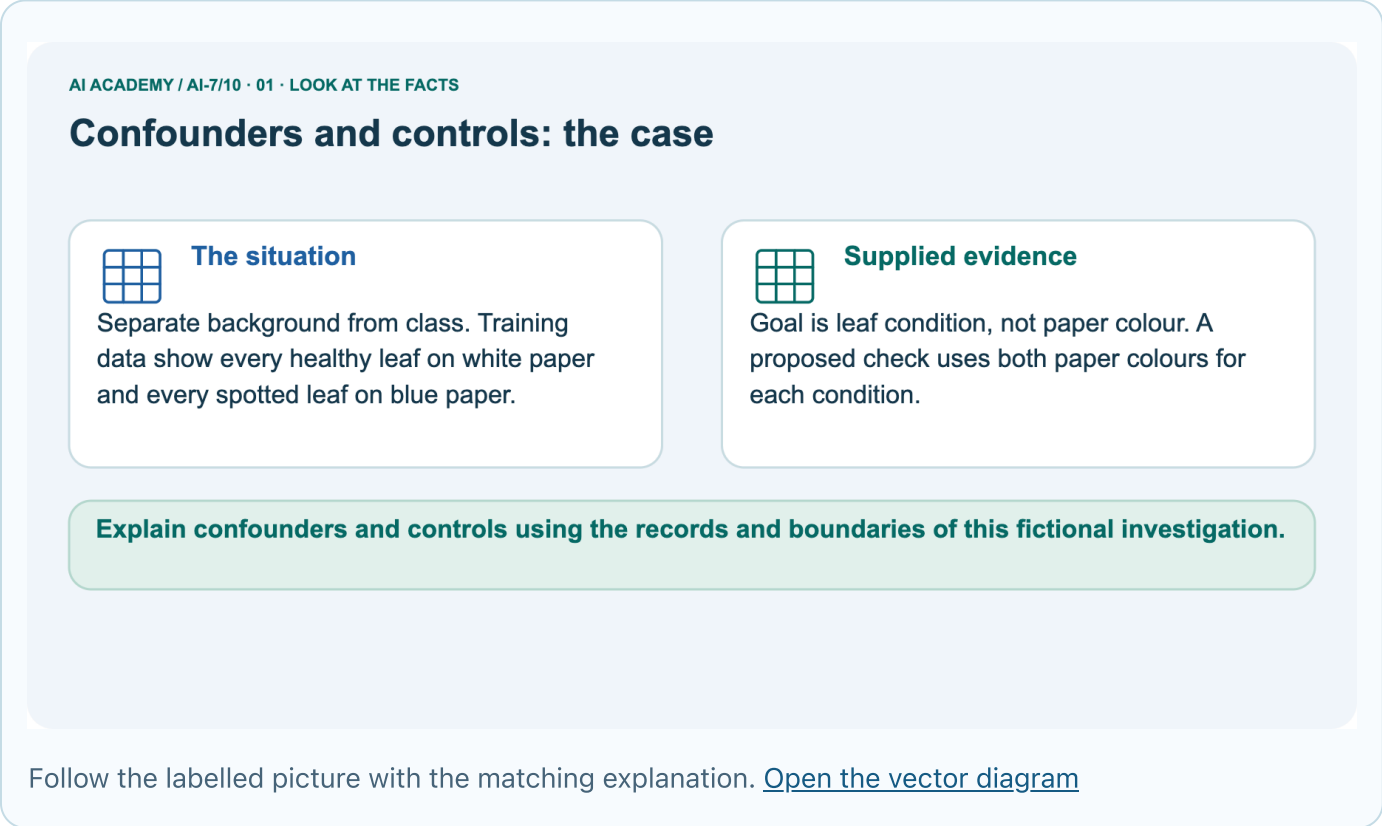
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Confounders and controls: the case



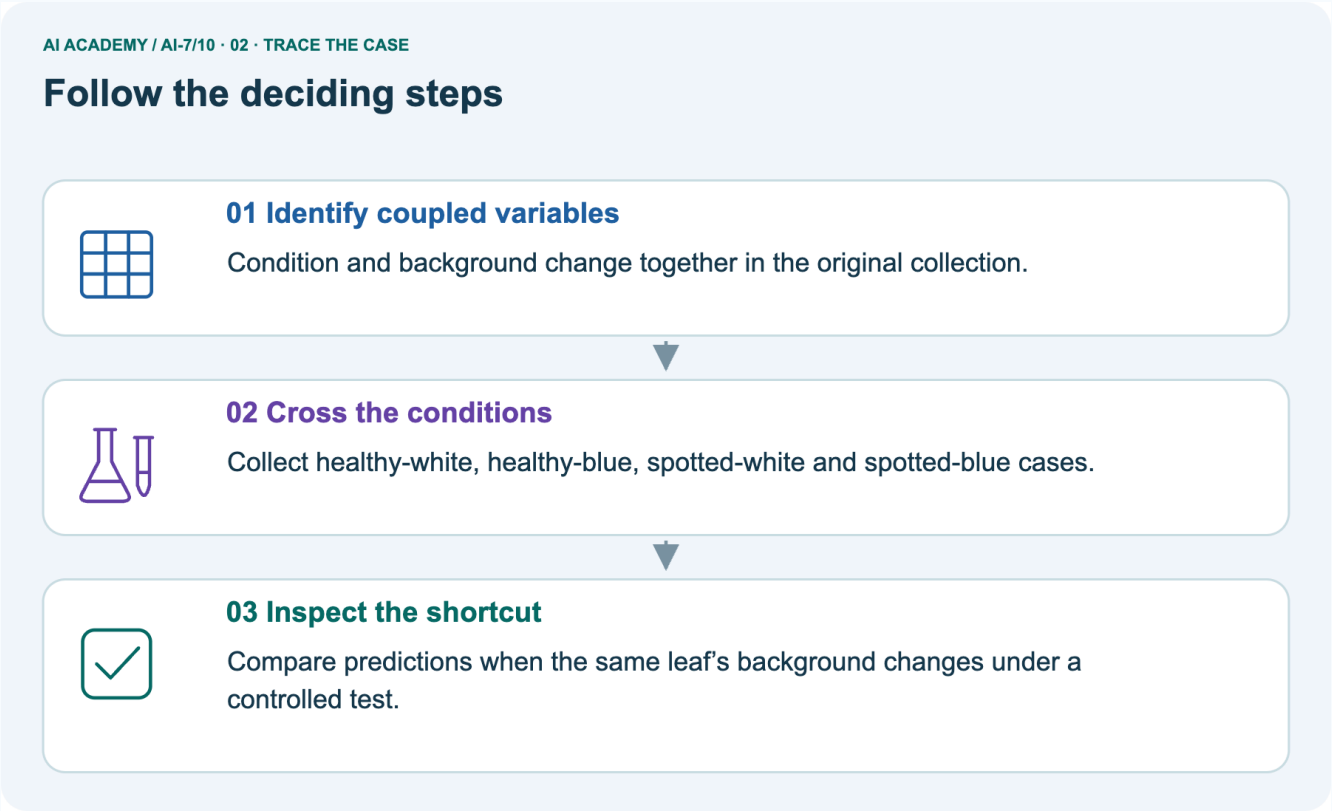
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper. Supplied facts: Goal is leaf condition, not paper colour.

A proposed check uses both paper colours for each condition.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Identify coupled variables: Condition and background change together in the original collection. → Cross the conditions: Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases. → Inspect the shortcut: Compare **prediction**s when the same leaf's background changes under a **control**led test.

Point to each numbered stage and explain which recorded **input** it uses.

03. Cross class with paper background

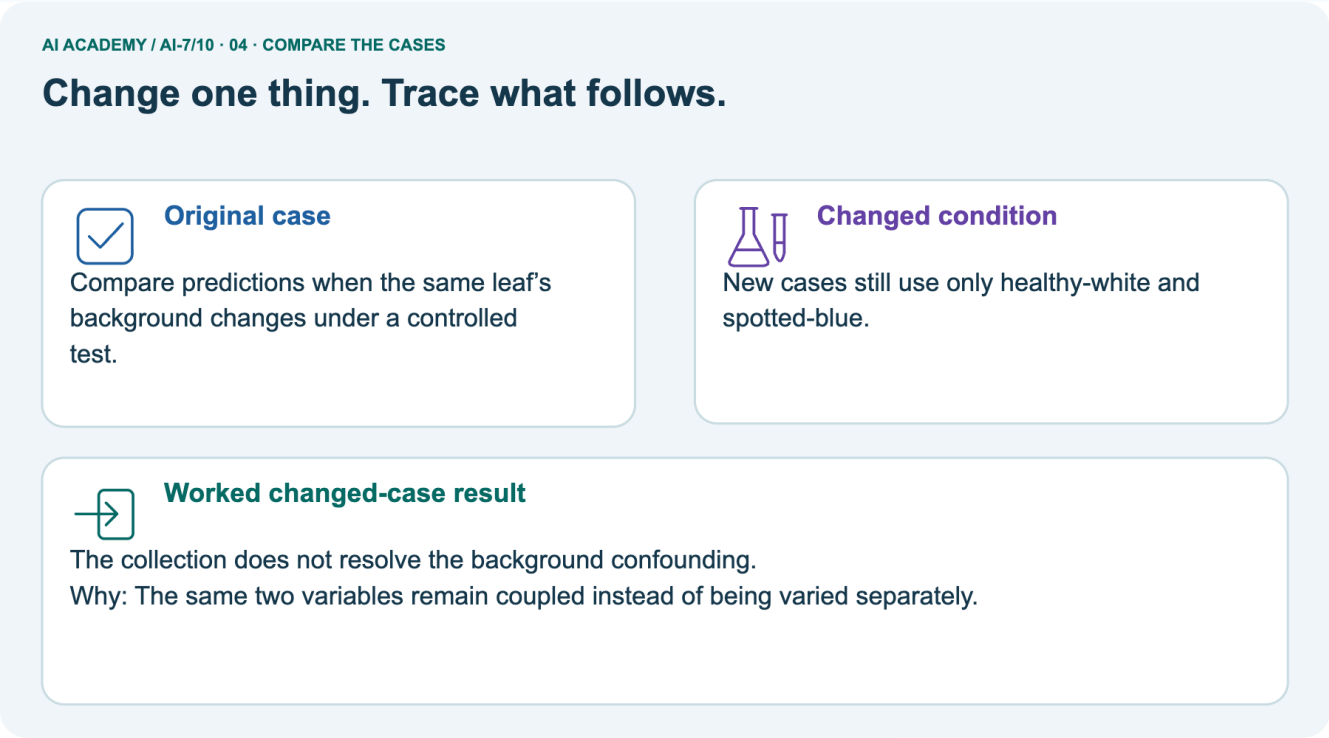


Follow the labelled picture with the matching explanation. Open the vector diagram

Cross class with paper background. Healthy: White paper = Collect, Blue paper = Collect. Spotted: White paper = Collect, Blue paper = Collect.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



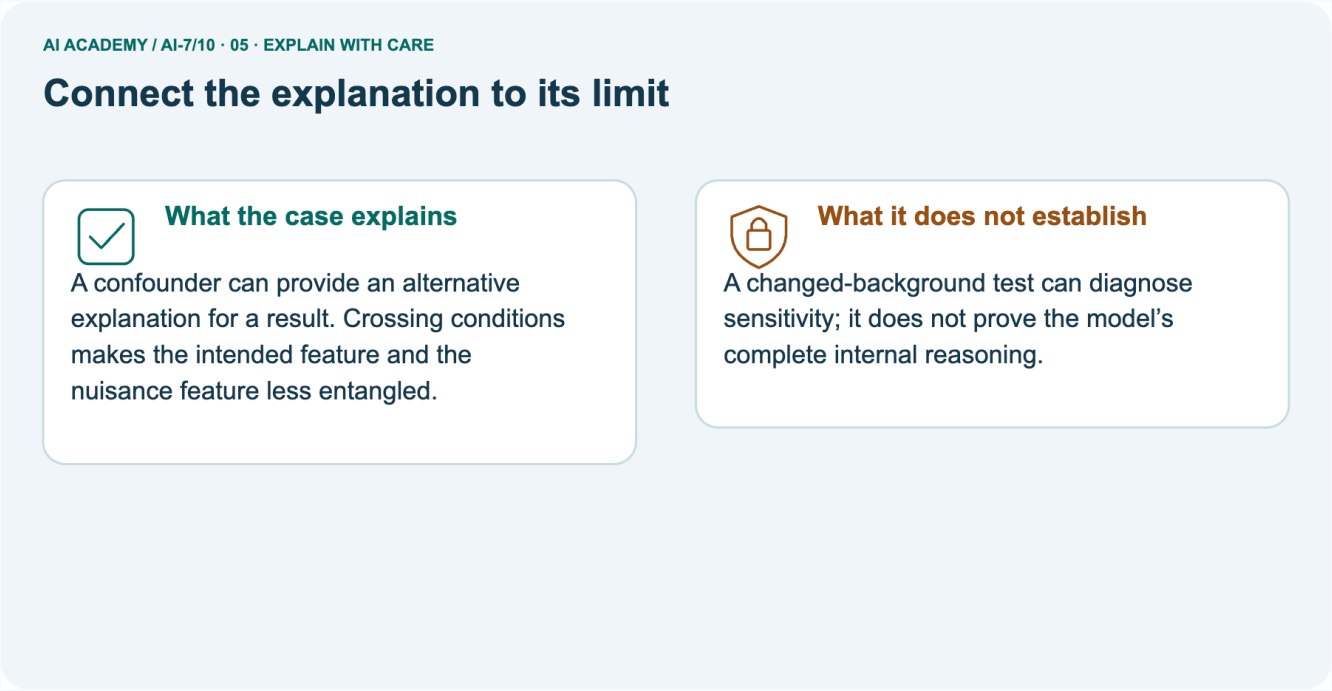
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare **prediction**s when the same leaf's background changes under a **control**led test. Changed condition: New cases still use only healthy-white and spotted-blue.

Expected result: The collection does not resolve the background confounding. Reason: The same two variables remain coupled instead of being varied separately.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



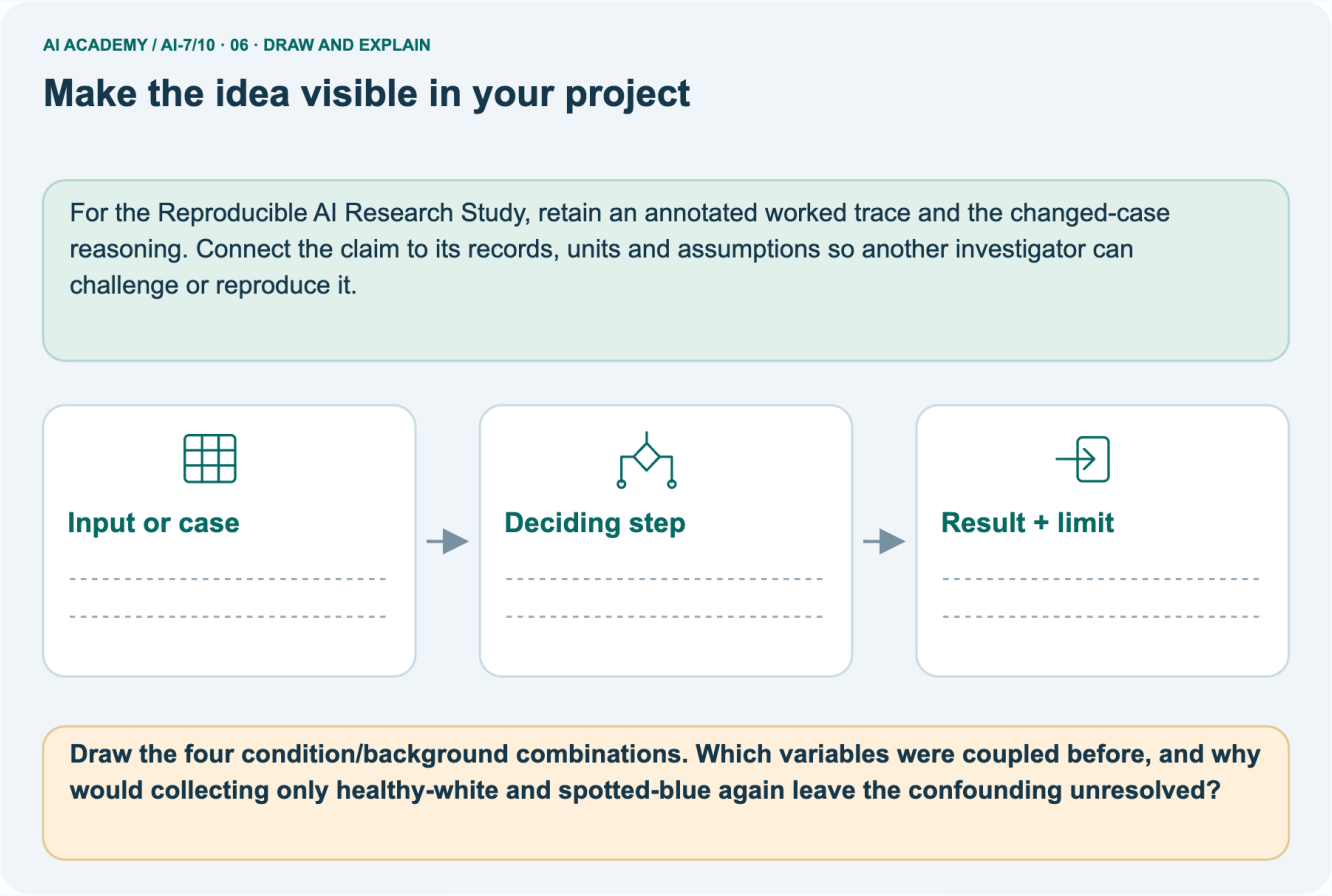
Follow the labelled picture with the matching explanation. Open the vector diagram

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Limit: A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Confounders and controls: the case

02. Follow the deciding steps

03. Cross class with paper background

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.